Regional Income Growth and House Price Dynamics in the Netherlands

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Set-up your environment

Regional Income Growth and House Price Dynamics in the Netherlands

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Tutorial group 3.3

Part 1 - Identify a Social Problem

Use APA referencing throughout your document. Here's a link to some explanation.

1.1 Describe the Social Problem

The Dutch housing market has been under pressure for several years now. Between 2013 and 2023 the average price of houses has more than doubled, while the income growth in the same period considerably lagged behind. The increasing gap between housing costs and purchasing power is not only an economic phenomenon, but also a social problem; it determines who does have access to a stable living situation and who doesn't.

It's striking that this development differs significantly across regions in the Netherlands. In the Randstad, prices increased much more than in the peripheral regions like Groningen or Zeeland, while the income development per region differs significantly too. Factors such as access to big employers, education level and local economic structure play an important role in this shift. And it is because of this regional variety that it's interesting to investigate the relation between income and house prices.

This project investigates the relationship between regional income growth and house price dynamics in Dutch provinces, based on CBS-data and the programming tool R. The goal is to discover in which regions the housing market detached most from the local income development.

Part 2 - Data Sourcing

2.1 Load in the data

For the income data we used nine separate CBS files, one for each year from 2015 to 2023. After cleaning each file individually (removing empty helper columns and adding a `jaar` year column), the files were combined row-wise into a single dataset with `rbind()`, ending in `datacombined15tot23 <- rbind(Data15161718, data2019, data2020, data2021, data2022, data2023)`. This income dataset was then merged with the house-price data (per province) and the population-density data, so that every observation is a province in a given year.

joining

2.2 Provide a short summary of the dataset(s)

For this study three datasets from the Centraal Bureau voor de Statistiek (CBS) were used. The first dataset, Bestaande koopwoningen; verkoopprijzen prijsindex 2020=100 (85773NED), contains data on the variables ‘gemiddelde verkoopprijs van bestaande koopwoningen en bijbehorende prijsindex op provinciaal niveau’. The second dataset, inkomen van huishoudens; huishoudenskenmerken, regio (86004NED), contains information on the ‘gemiddelde inkomen van huishoudens, op provinciaal niveau’. Lastly, the third dataset, bevolkingsdichtheid op 1 januari(70072NED), contains information on the population density in the Netherlands.

All datasets are available on CBS StatLine. The dataset on the prices of houses spans from 2015 until 2025, the dataset on income only covers 2015 until 2023 (In order to see other years or regions, the filter option should be used to select the desired years). The dataset on population density in the Netherlands spans from 1995 until 2025. For the analysis in this research the overlapping years of 2015-2023(or a part thereof) are studied.

2.3 Describe the type of variables included

The datasets mostly have social economic variables (SES). The house price dataset measures the average sale price of existing houses and the price index of these sales. This is an indicator of the accessibility of the housing market in the Netherlands. The dataset on the income of households measures the average income of households per province, as a benchmark for wealth in that region. The dataset on population density, again, is based on administrative data. It measures the amount of inhabitants per square kilometer in every province.

The data originates from administrative sources, not interviews or questionnaires. The average house prices are based on the ‘Basisregistratie Kadaster’, in which every official sale of a house is recorded by law. The data on the income of households are deduced from tax returns and benefit registrations, processed by the CBS into regional stats. The data on population density is computed by dividing the amount of inhabitants officially listed by the amount of square kilometers in the region.

Administrative data mostly doesn’t suffer from non-responses and measurement errors. A possible challenge is the fact that administrative definitions change over time; income can be defined differently from one year onto the next. This could affect the comparability over the years.

Part 3 - Quantifying

3.1 Data cleaning

Data cleaning datacombined15tot23goed. we mainly used this code to clean this data set: names(datacombined15tot23goed)[na
== “X1.000.euro. . . 6”] <- “gem_st_inkomen_x_100” ETC.

Data cleaning woningprijsprov: we mainly used this code to clean this data set: woningprijsprov‘1995‘ < $-format(woningprijsprov$`1995`, big.mark = “.”, scientific = FALSE)

Please use a separate ‘R block’ of code for each type of cleaning. So, e.g. one for missing values, a new one for removing unnecessary variables etc.

3.2 Generate necessary variables

Variable 1 – average disposable income per province Our first generated variable is the average disposable household income per province, averaged over the years 2015–2023. We created it by aggregating the cleaned income dataset (`datacombined15tot23goed`) with `aggregate()`, taking the mean of `gem_bstb_inkomen_x_100` for each province. The result (`gem_inkomen_provincie`) gives one income figure per province and serves as a stable benchmark of regional purchasing power, which we later compare against house prices. The code used was: gem_inkomen_provincie <- aggregate(gem_bstb_inkomen_x_100 ~ Provincies, data = datacombined15tot23goed, FUN = mean)

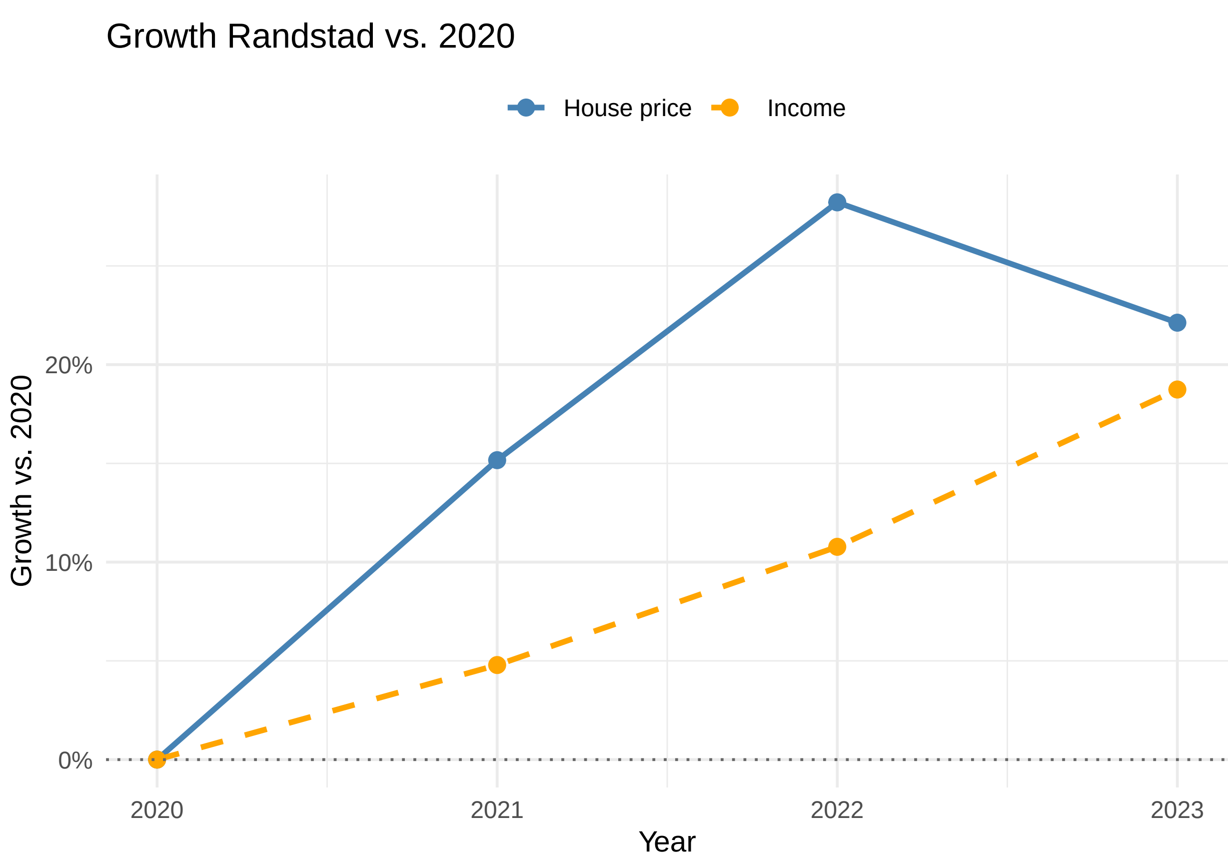
Variable 2 – Randstad versus outside the Randstad For the second variable we grouped the provinces into two regions, the Randstad (including Flevoland) and the rest of the country, and computed the average income and average house price for each region per year. This required reshaping the house-price dataset from wide to long format (one row per province per year) and merging it with the income data, so that income and house prices could be compared on the same regional and yearly basis. This variable lets us contrast the economically dense Randstad with the peripheral provinces.

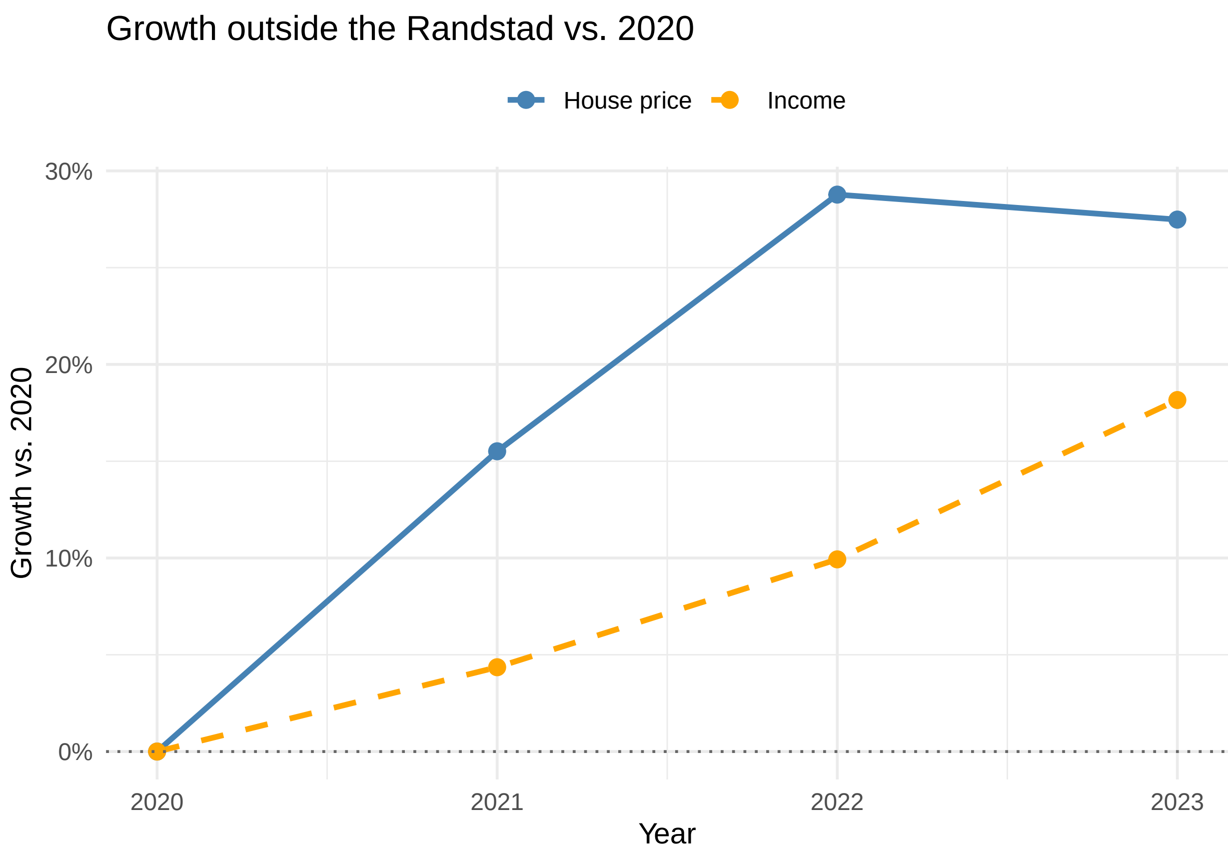
Variable 3 – population density and house price per province The third variable combines population density with the average house price for each province per year (2020–2023). We reshaped both the density data and the house-price data to long format and merged them on province and year, producing `bevolkingsdichtheid_en_huizenprijzen_per_provincie`. This variable allows us to test whether more densely populated provinces also have higher house prices (used in section 3.5). A fourth variable, the price-to-income ratio, is generated later in the event analysis (section 3.6).

3.3 Visualize temporal variation

```r
library(tidyverse)
library(scales)

df <- read.csv("goede data/randstad_vs_buiten_randstad.csv", stringsAsFactors = FALSE)

df_plot <- df %>%
  arrange(regio, jaar) %>%
  group_by(regio) %>%
  mutate(
    groei_huizen = (gem_huizenprijs / first(gem_huizenprijs)) - 1,
    groei_inkomen = (gem_bstb_inkomen_euro / first(gem_bstb_inkomen_euro)) - 1
  ) %>%
  ungroup()

df_long <- df_plot %>%
  pivot_longer(cols = c(groei_huizen, groei_inkomen),
               names_to = "type",
               values_to = "groei") %>%
  mutate(type = recode(type,
                       "groei_huizen" = "House price",
                       "groei_inkomen" = "Income"))

maak_grafiek <- function(data, titel) {
  ggplot(data, aes(x = jaar, y = groei, color = type, linetype = type)) +
    geom_line(linewidth = 1) +
    geom_point(size = 2.5) +
    geom_hline(yintercept = 0, linetype = "dotted", color = "grey40") +
```

```
    scale_y_continuous(labels = percent_format(accuracy = 1)) +
    scale_x_continuous(breaks = c(2020, 2021, 2022, 2023)) +
    scale_color_manual(values = c("House price" = "steelblue", "Income" = "orange")) +
    scale_linetype_manual(values = c("House price" = "solid", "Income" = "dashed")) +
    labs(title = titel, x = "Year", y = "Growth vs. 2020",
         color = "", linetype = "") +
    theme_minimal() +
    theme(legend.position = "top")
}

grafieken_tonen <- function() {
  print(maak_grafiek(
    df_long %>% filter(regio == "Randstad (incl. Flevoland)"),
    "Growth Randstad vs. 2020"
  ))
  print(maak_grafiek(
    df_long %>% filter(regio == "Buiten Randstad"),
    "Growth outside the Randstad vs. 2020"
  ))
}

grafieken_tonen()
```



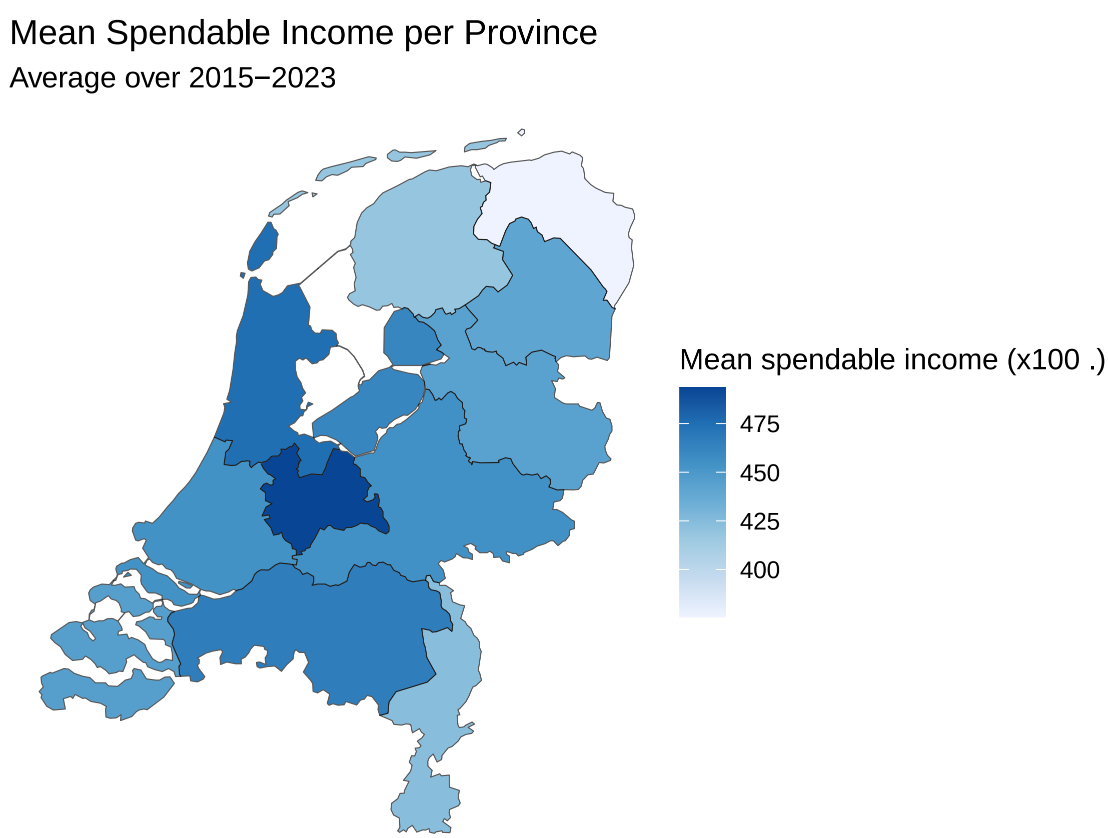


These two graphs show the growth of house prices and disposable income relative to 2020, for the Randstad and for the rest of the country. In both regions house prices rose much faster than income: prices climbed by roughly 15% in 2021 and another 11% in 2022, while income grew by only about 4% to 5% per year. The gap between the two lines – the widening distance between housing costs and what people earn – is exactly the affordability problem central to this report, and it is visibly larger in the Randstad than outside it.

3.4 Visualize spatial variation

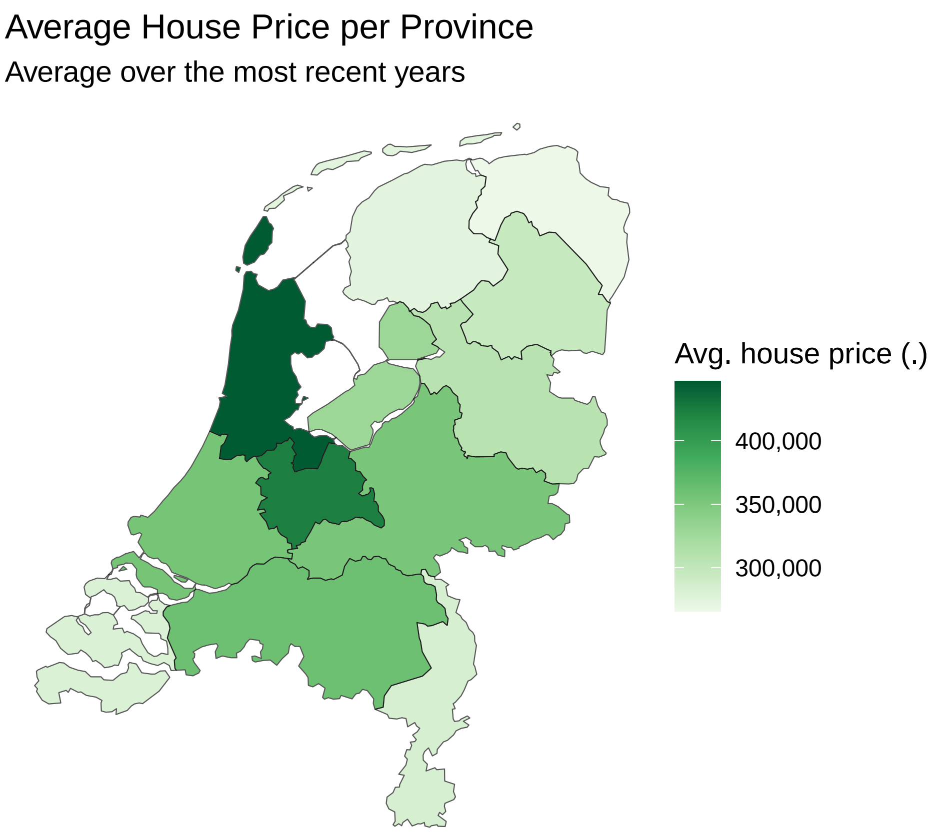
```
require(sf)
require(ggplot2)
nl_map <- st_read("https://cartomap.github.io/nl/wgs84/provincie_2024.geojson", quiet = TRUE)

my_data <- data.frame(
  Provincies = c("Groningen", "Fryslân", "Limburg", "Drenthe", "Overijssel",
                 "Zeeland", "Zuid-Holland", "Gelderland", "Flevoland",
                 "Noord-Brabant", "Noord-Holland", "Utrecht"),
  mean_income = c(375.5556, 418.3333, 423.7778, 440.7778, 443.6667,
                  445.1111, 453.8889, 454.7778, 461.2222, 466.7778,
                  474.8889, 493.5556)
)
nl_map <- merge(nl_map, my_data, by.x = "statnaam", by.y = "Provincies")
ggplot(nl_map) +
  geom_sf(aes(fill = mean_income)) +
  scale_fill_distiller(
```

```r
    palette = "Blues",
    direction = 1,         # 1 = darker means higher value
    name = "Mean spendable income (x100 €)"
  ) +
  theme_void() +
  labs(
    title = "Mean Spendable Income per Province",
    subtitle = "Average over 2015-2023"
  )
```




```r
nl_map <- st_read("https://cartomap.github.io/nl/wgs84/provincie_2024.geojson", quiet = TRUE)


my_data2 <- data.frame(
  Provincies = c("Groningen", "Fryslân", "Drenthe", "Overijssel", "Flevoland",
                 "Gelderland", "Utrecht", "Noord-Holland", "Zuid-Holland",
                 "Zeeland", "Noord-Brabant", "Limburg"),
  huizenprijs = c(265840.0, 274763.0, 296620.4, 307718.0, 329630.8,
                  353163.8, 423367.6, 446549.2, 355059.0,
                  280299.6, 360750.2, 283216.8)
)

nl_map <- merge(nl_map, my_data2, by.x = "statnaam", by.y = "Provincies")

ggplot(nl_map) +
```

```r
  geom_sf(aes(fill = huizenprijs)) +
  scale_fill_distiller(
    palette = "Greens",
    direction = 1,
    name = "Avg. house price (€)",
    labels = scales::comma
  ) +
  theme_void() +
  labs(
    title = "Average House Price per Province",
    subtitle = "Average over the most recent years"
  )
```

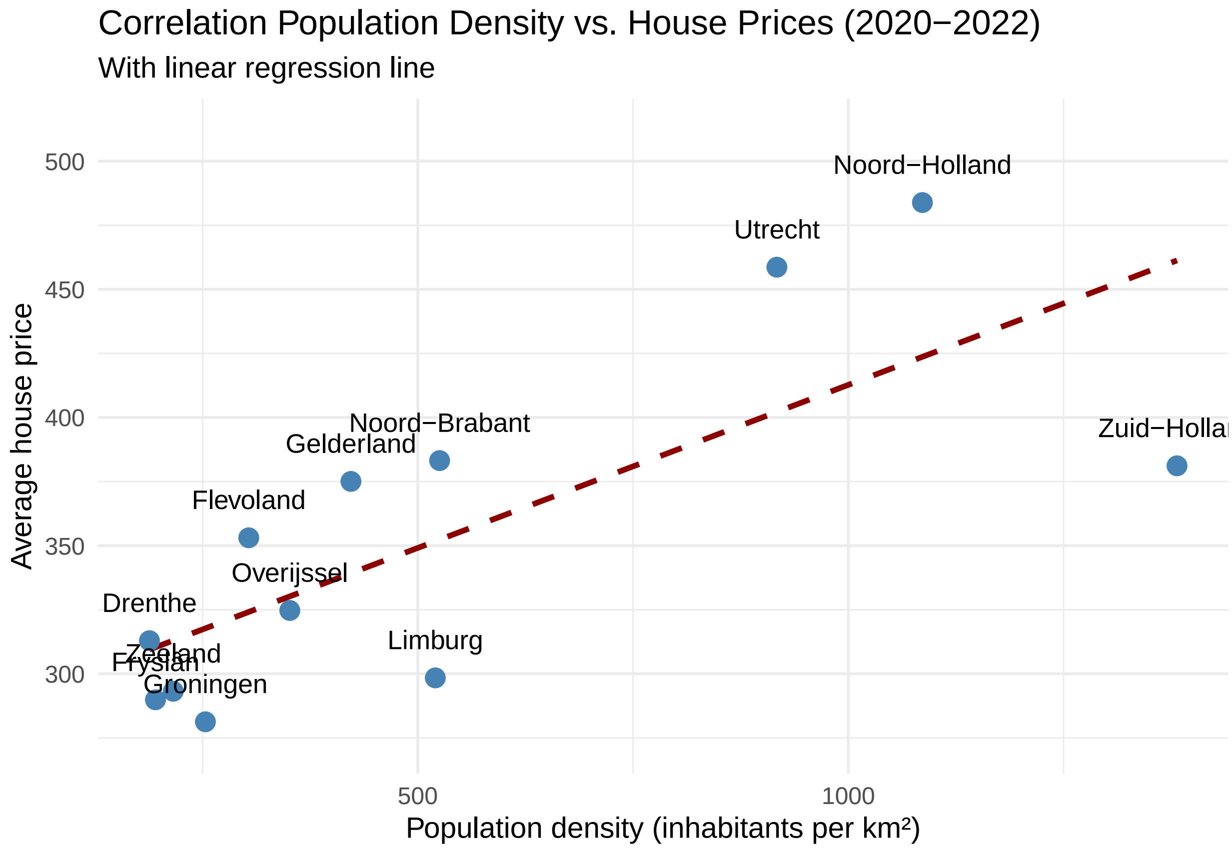


The two maps show the spatial distribution of, respectively, average disposable income and the average house price per province. The highest incomes and the highest house prices both concentrate in the same central provinces (Utrecht and Noord-Holland), while the peripheral provinces (Groningen, Zeeland, Drenthe) score low on both. However, house prices rise more steeply than income across the map, so the regions that are already expensive become relatively even less accessible. This spatial inequality in access to housing is the core of our social problem.

3.5 Visualize sub-population variation

```r
library(tidyverse)
```

```r
df_provincies <- read.csv("goede data/bevolkingsdichtheid_en_huizenprijzen_per_provincie.csv", stringsAs

df_plot <- df_provincies %>%
  mutate(Provincies = str_remove(Provincies, " \\(PV\\)")) %>%
  group_by(Provincies) %>%
  summarise(
    dichtheid = mean(bevolkingsdichtheid_per_provincie, na.rm = TRUE),
    huizenprijs = mean(gem_huizenprijs, na.rm = TRUE)
  ) %>%
  drop_na(dichtheid, huizenprijs)

ggplot(data = df_plot, aes(x = dichtheid, y = huizenprijs, label = Provincies)) +
  geom_smooth(method = "lm", formula = y ~ x, color = "darkred", linetype = "dashed", se = FALSE) +
  geom_point(color = "steelblue", size = 3) +
  geom_text(vjust = -1.5, size = 3.5) +
  scale_y_continuous(expand = expansion(mult = c(0.1, 0.2))) +
  labs(
    title = "Correlation Population Density vs. House Prices (2020-2022)",
    subtitle = "With linear regression line",
    x = "Population density (inhabitants per km²)",
    y = "Average house price"
  ) +
  theme_minimal()
```

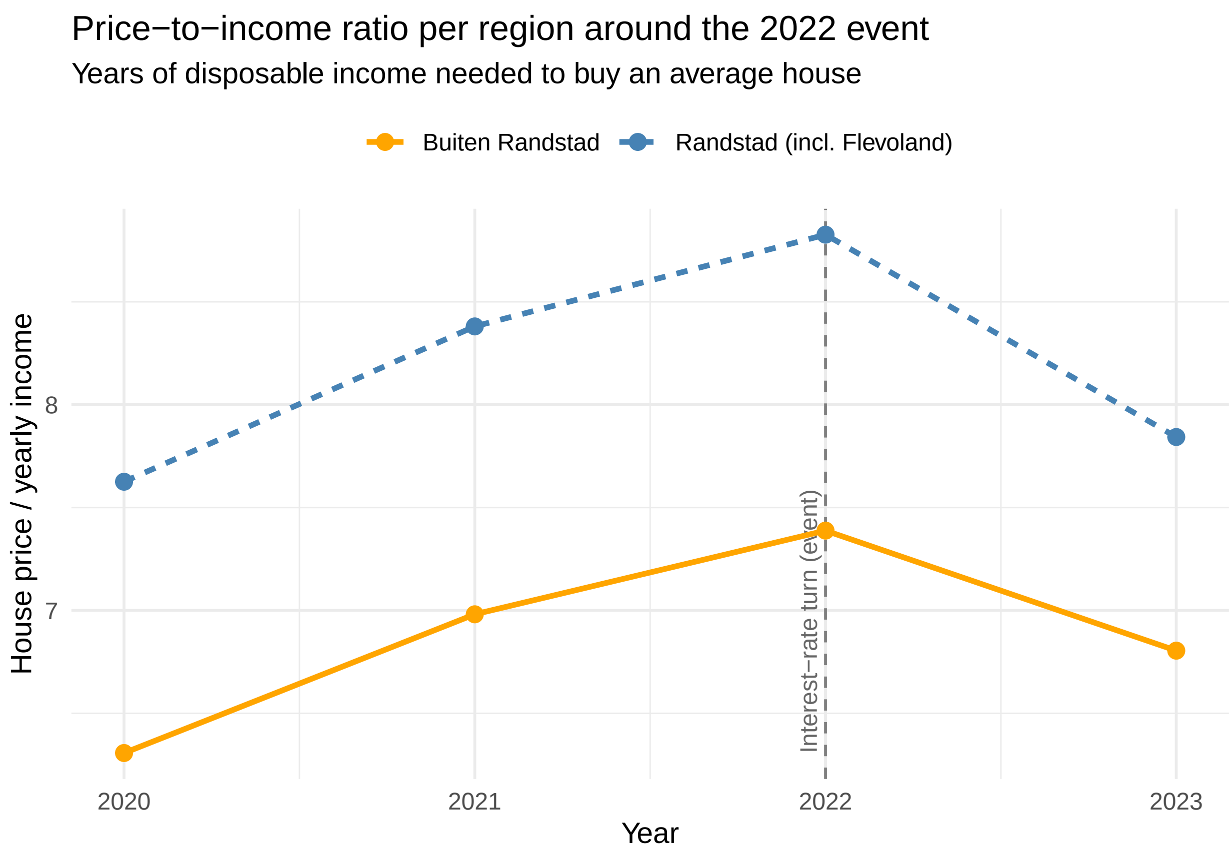


This scatter plot relates population density to the average house price per province, with a linear regression line added. The upward slope shows a clear positive association: more densely populated provinces tend to have higher house prices. This is relevant to our social problem because it indicates that scarcity of space, not just income, drives prices up – precisely in the crowded provinces where most people want or need to live, housing is least affordable.

3.6 Event analysis

For the event analysis we study the relationship between two variables – average disposable household income and the average house price – around a clear event: the turn in mortgage interest rates in 2022. To do this we generate a fourth variable, the **price-to-income ratio** (average house price divided by average yearly disposable income), which can be read as "the number of years of income needed to buy an average house". The plot below shows this ratio over time for the Randstad and the rest of the country, with the 2022 event marked.

```r
library(tidyverse)

# New variable (variable 4): price-to-income ratio = average house price (in euro)
# divided by the average yearly disposable income (in euro). We use the regional
# dataset, which already contains both income (in euro) and the house price (x1000 euro).
ratio_data <- read.csv("goede data/randstad_vs_buiten_randstad.csv",
                       stringsAsFactors = FALSE)

ratio_data <- ratio_data %>%
  mutate(
    huizenprijs_euro = gem_huizenprijs * 1000,
    verhouding_prijs_inkomen = huizenprijs_euro / gem_bstb_inkomen_euro
  )

write.csv(ratio_data, "goede data/verhouding_prijs_inkomen.csv", row.names = FALSE)

# Event association plot: the ratio over time, with the 2022 interest-rate turn marked.
ggplot(ratio_data, aes(x = jaar, y = verhouding_prijs_inkomen,
                       color = regio, linetype = regio)) +
  geom_vline(xintercept = 2022, linetype = "dashed", color = "grey50") +
  annotate("text", x = 2022, y = min(ratio_data$verhouding_prijs_inkomen),
           label = "Interest-rate turn (event)", angle = 90,
           vjust = -0.5, hjust = 0, size = 3, color = "grey40") +
  geom_line(linewidth = 1) +
  geom_point(size = 2.5) +
  scale_x_continuous(breaks = c(2020, 2021, 2022, 2023)) +
  scale_color_manual(values = c("Randstad (incl. Flevoland)" = "steelblue",
                                "Buiten Randstad" = "orange")) +
  labs(
    title = "Price-to-income ratio per region around the 2022 event",
    subtitle = "Years of disposable income needed to buy an average house",
    x = "Year", y = "House price / yearly income",
    color = "", linetype = ""
  ) +
  theme_minimal() +
  theme(legend.position = "top")
```




```
# Alternative (swap in if you prefer a classic association scatter instead of the
# time/event view): per-province average income on the x-axis and average house price
# on the y-axis with a regression line.
# inc <- read.csv("goede data/datacombined15tot23goed.csv")
# hp  <- read.csv("goede data/bevolkingsdichtheid_en_huizenprijzen_per_provincie.csv")
# assoc <- merge(aggregate(gem_bstb_inkomen_x_100 ~ Provincies, inc, mean),
#                aggregate(gem_huizenprijs ~ Provincies, hp, mean), by = "Provincies")
# ggplot(assoc, aes(gem_bstb_inkomen_x_100 * 100, gem_huizenprijs * 1000)) +
#   geom_smooth(method = "lm", se = FALSE) + geom_point()
```

This event analysis is central to our social problem. Between 2020 and 2022 the price-to-income ratio rose in both regions – from about 7.6 to 8.8 years of income in the Randstad and from 6.3 to 7.4 outside it – meaning houses became steadily less affordable relative to local income. After the 2022 interest-rate turn the ratio falls again in 2023 (prices declined by roughly 1% to 5% while incomes kept rising by about 7%). The fact that affordability moves with the financing event rather than with local income growth suggests that house prices in this period were driven more by mortgage conditions than by regional purchasing power – and that the affordability gap is structurally largest in the Randstad.

Part 4 - Discussion

4.1 Discuss your findings

Our analysis points to one consistent pattern: across Dutch provinces, house prices have grown considerably faster than disposable income, so housing has become less affordable almost everywhere. The decoupling is strongest in the Randstad, where the price-to-income ratio is both the highest and the fastest-rising, but the peripheral provinces follow the same direction. The spatial analysis shows that high prices cluster in the same densely populated, high-income provinces, yet because prices outrun income, even relatively wealthy regions face an affordability gap.

The event analysis adds an important nuance. The affordability ratio peaks in 2022 and then falls in 2023, mirroring the turn in mortgage interest rates rather than any change in local income. This suggests that the recent price dynamics were driven more by financing conditions than by regional purchasing power, which means affordability could deteriorate again quickly if borrowing becomes cheaper.

Several limitations should be kept in mind. The overlapping window in which we have both income and house-price data per province is short (2020–2023), so the trends rest on few observations and should be read as indicative rather than conclusive. Administrative income definitions can change between years, which may affect comparability, and the analysis covers only the twelve provinces, so within-province differences (e.g. between cities and rural areas) are hidden. Finally, our results show association, not causation: income and house prices move together, but both are likely influenced by common drivers such as interest rates and the supply of housing.

Part 5 - Reproducibility

5.1 Github repository link

Provide the link to your PUBLIC repository here:

5.2 Reference list

Centraal Bureau voor de Statistiek (2024). Inkomen van huishoudens; huishoudenskenmerken, regio [Dataset]. CBS. https://opendata.cbs.nl/#/CBS/nl/dataset/86004NED/table

Centraal Bureau voor de Statistiek. (2026). Bestaande koopwoningen; verkoopprijzen prijsindex 2020=100 [Dataset]. CBS. https://opendata.cbs.nl/#/CBS/nl/dataset/85773NED/table

Centraal Bureau voor de Statistiek. (2026). Regionale kerncijfers Nederland [Dataset].CBS. https://opendata.cbs.nl/#/CBS/nl/dataset/70072ned/table?dl=17F53